Lecture 2

Building Blocks of Electronics

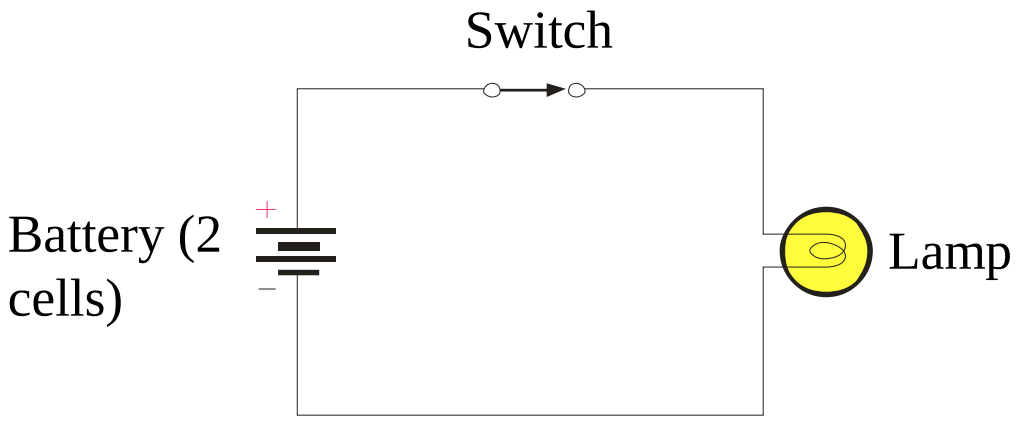
Sonia Chacko, PhD.

Outline

- Introduction to Electronics
- Current
- Resistance

What is Electronics?

- Electronics is the study and use of devices that control the flow of electrical current.
- Deals with signals, circuits, and components.

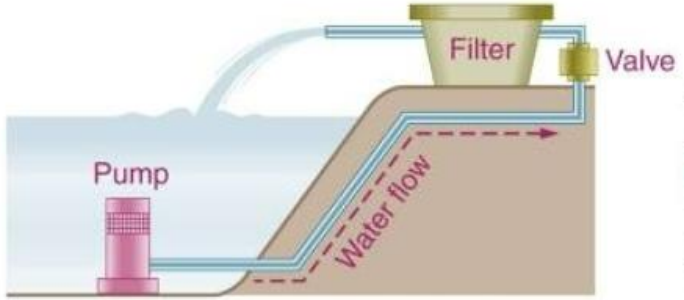


A simple electric circuit

Basic Electrical Quantities

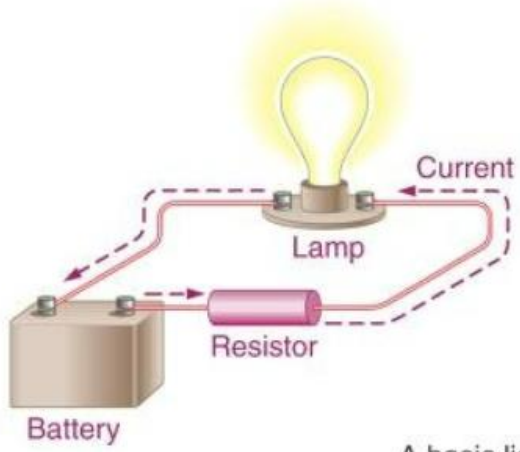
- **Voltage (V):** Electrical “pressure” (push).
- **Current (I):** Flow of electrons (like water flow).
- **Resistance (R):** Opposition to flow.
- Power (P): Energy used (P = V × I).

The Oldest Analogy in Electronics



The **pump** provides the **hydraulic force** that pushes **water** through the **pipe**. The rate of **water flow** depends on the pressure provided by the pump, and cannot exceed the capacity of the pipe. The **valve** is used to set the water flow to some desired rate.

A water garden filtration system



The **battery** provides the **electromotive force (EMF)** that pushes **charge** through the **wire**. The **current** depends on the EMF provided by the battery, and cannot exceed the capacity of the wire. The **resistor** is used to set the current to some desired rate.

A basic lighting circuit

Ohm’s Law

The most important fundamental law in electronics is **Ohm’s law**, which relates voltage, current, and resistance.

Formula: **V = I × R**

- *Current is **directly** proportional to **voltage***
- *Current is **inversely** proportional to **resistance***

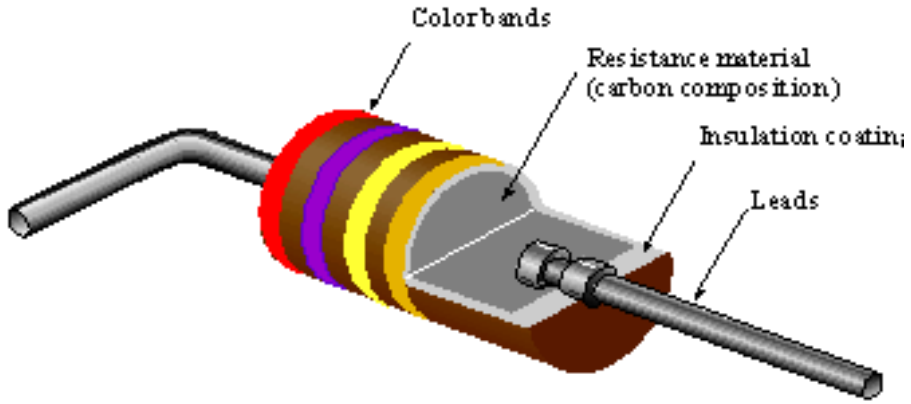
$$V = IR$$

$$I = \frac{V}{R}$$

Resistance

- Resistance – opposition to current
- Unit of Resistance – Ohm (Ω - Greek letter omega)
- Ohm – the **amount of resistance** that **limits current** to **one ampere** **when one volt is applied**
- One ohm (1 Ω) is the resistance exits when there is one ampere (1 A) is in a material with one volt (1 V) is applied across the material.

Components designed to have a specific amount of resistance are called resistors.



Basic Electrical Quantities - Units

Electrical Property	Units	Description
Current (I)	Ampere (A)	The rate of flow of electrical charge
Voltage (E or V)	Volts (V)	Electric Force
Resistance (R)	ohm (Ω)	The opposition to current.

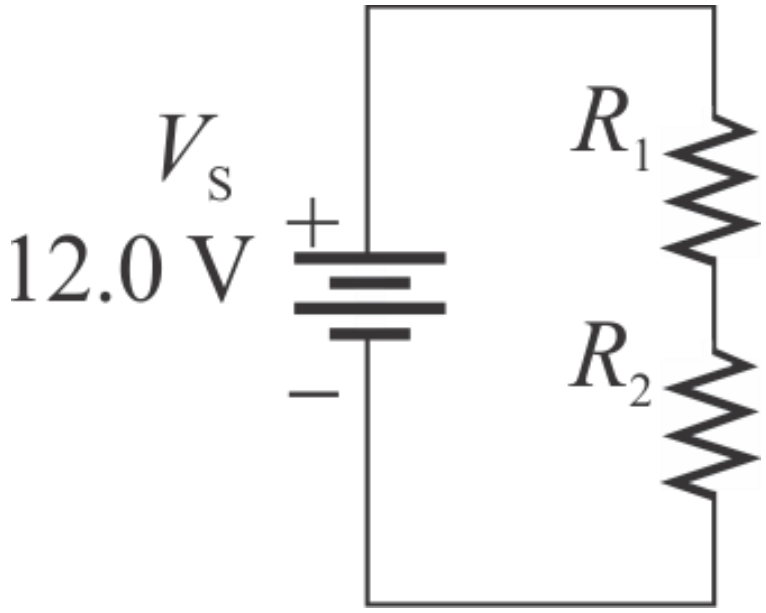
Basic Components

- **Resistor:** Controls current.
- **LED:** Light Emitting Diode.

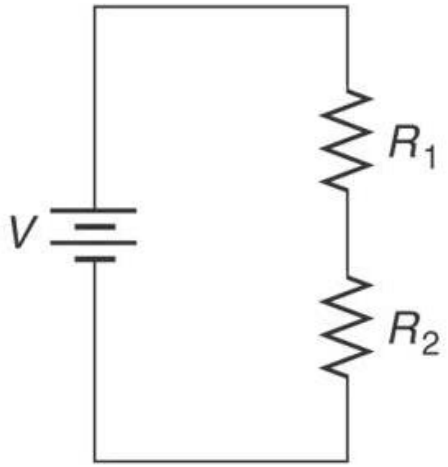
Lab Exercises - Resistors

- Series Circuits
- Parallel Circuits
- Voltage Divider

Series Resistor Circuits



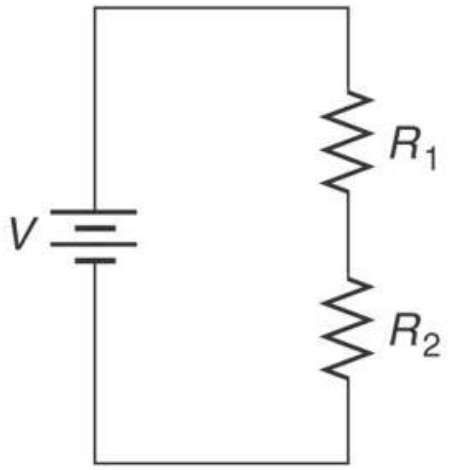
Series Resistor Circuits



Observation & Discussion:

- Is the current the same through all resistors?
- What is the voltage across the resistor R1?
- What is the voltage across the resistor R2?
- Do the voltages across resistors add up to source voltage?
- What is the total resistance of the Circuit?

Series Resistor Circuits



- In series resistor circuit, current is same, voltage divides.

$$V_S = V_1 + V_2 + \ldots + V_n$$

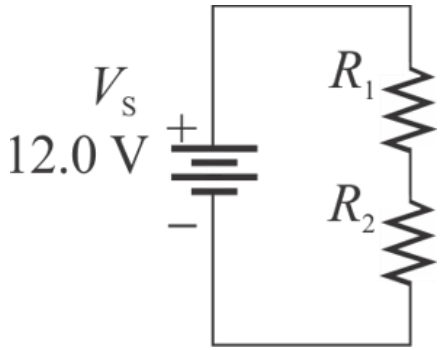
- Total Series Resistance is sum of individual resistances

$$R_T = R_1 + R_2 + \ldots + R_n$$

Where, R_T = the total circuit resistance

Voltage Divider Circuit

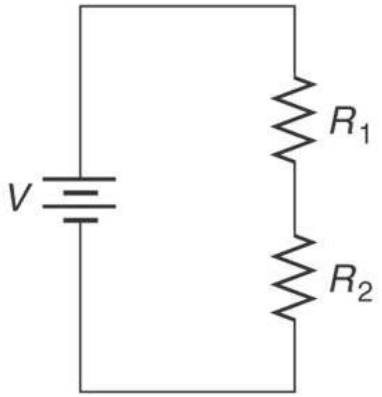
- The Voltage Divider Relationship
  - Voltage Divider – often used to describe a series circuit



Voltage Divider Circuit

Voltage Divider Rule:

The voltage drop across any given resistor in a series circuit is equal to the ratio of that resistor to the total resistance, multiplied by source voltage.



$$V_1 = \left(\frac{R_1}{R_T}\right) V_S$$

$$V_2 = \left(\frac{R_2}{R_T}\right) V_S$$

$$\frac{V_n}{V_S} = \frac{R_n}{R_T} \Rightarrow V_n = V_S \frac{R_n}{R_T}$$

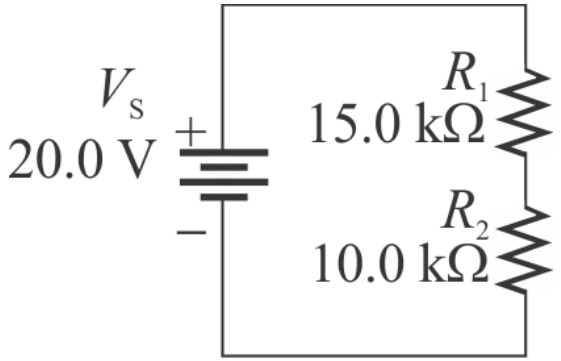
where

R_n = the resistor of interest

V_n = the voltage drop across R_n

(where n is the component number)

What is the voltage across R_2?



The total resistance is 25 kΩ.
Applying the voltage divider formula:

$$V_2 = \left(\frac{R_2}{R_T}\right) V_S = \left(\frac{10\ \text{k}\Omega}{25\ \text{k}\Omega}\right) 20\ \text{V} = \quad 8.0\ \text{V}$$

Notice that 40% of the source voltage is across R2, which represents 40% of the total resistance.

What is the voltage across R_1?

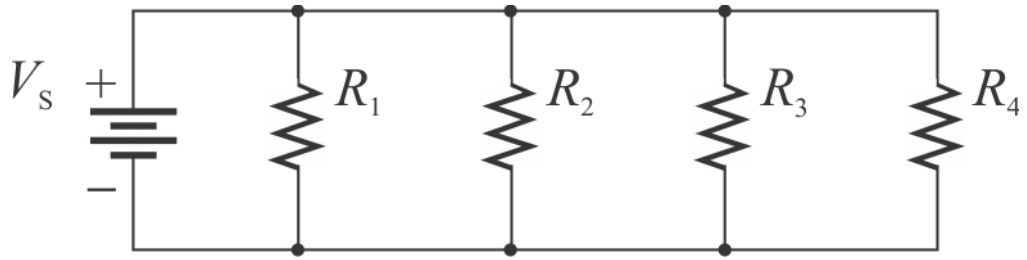
Apply KVL or Voltage divider rule:

KVL: $V_1 = Vs - V_2$

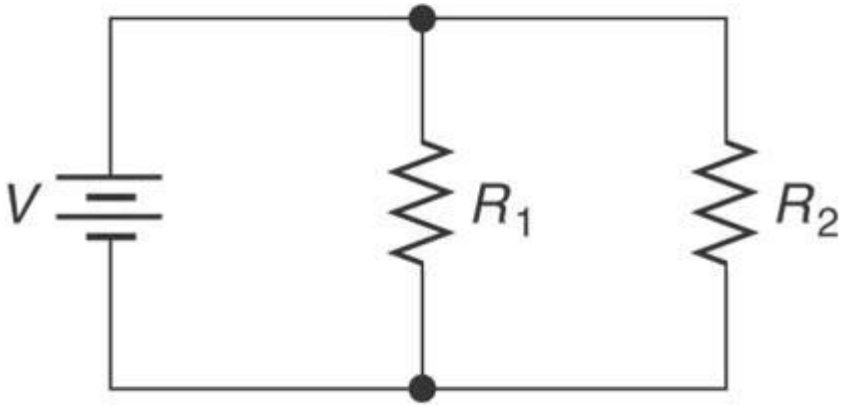
Resistors in Parallel

Resistors that are connected between the same two points are said to be in **parallel**.

A ***parallel circuit*** is identified by the fact that it has more than one current path (branch) connected to a common voltage source.

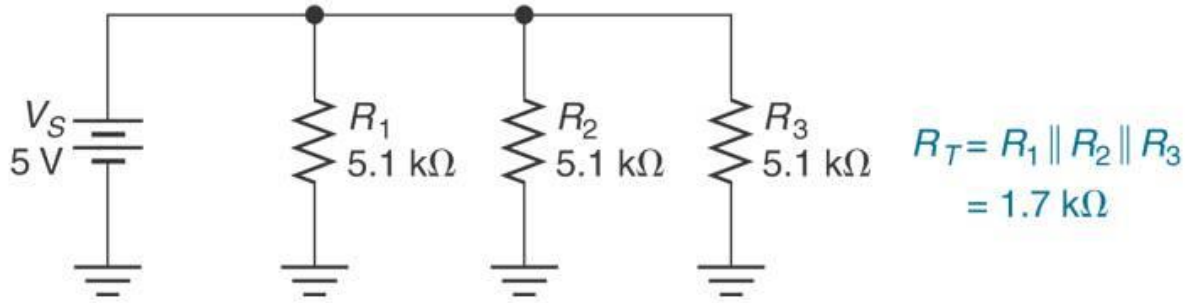


Parallel Resistor Circuits



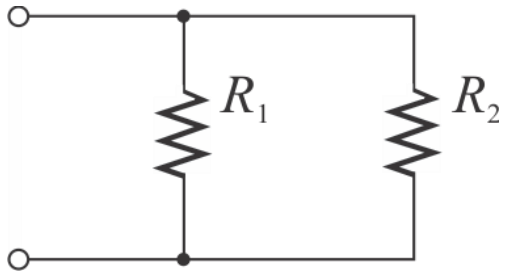
$$R_T = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \ldots + \frac{1}{R_n}}$$

- Calculating Total Resistance When Branch Resistances are Equal – Divide the value of one resistor by the number of branches



$$R_T = \frac{5.1\text{k}\Omega}{3} = 1.7\text{k}\Omega$$

Two Parallel Resistors



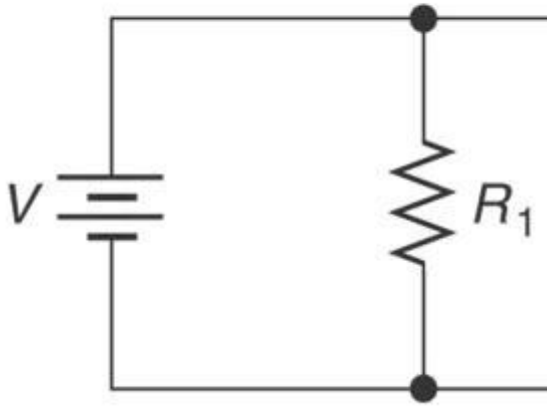
The special case of resistance of two parallel resistors can be found by:

$$R_T = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}}$$

which can be reduced to $R_T = \frac{R_1 R_2}{R_1 + R_2}$ ← This is called the product-over-sum rule.

Calculate the total resistance if R_1 = 27 kΩ and R_2 = 56 kΩ 18.2 kΩ

Power Dissipation in Resistors



- What is the power dissipation across the resistor?